Face2Health: A Multimodal Non-Invasive Health Risk Screening Pipeline Using Vision Transformer and Monotonic Calibrated XGBoost

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***Abstract*—This paper presents Face2Health, a cascading multimodal machine learning architecture engineered for non-invasive, accessible screening of systemic non-communicable diseases (NCDs). The pipeline sequentially couples: (1) a Vision Transformer (ViT-H/14) fine-tuned for Body Mass Index (BMI) prediction with 25-pass Monte Carlo Dropout for epistemic uncertainty quantification alongside scale-invariant facial morphometrics; (2) a tabular Extreme Gradient Boosting (XGBoost) regressor mapping estimated BMI and physiological covariates to Dual-Energy X-ray Absorptiometry (DEXA)-derived Total Body Fat Percentage; and (3) monotonically regularized, Platt-calibrated XGBoost classifiers estimating continuous posterior probabilities for Type 2 Diabetes Mellitus and Essential Hypertension. Benchmarked on the CDC NHANES adult cohort (N = 3,540 for Diabetes, N = 3,554 for Hypertension), this work resolves two pervasive failure modes in clinical machine learning: cascading error compounding and the 0-Recall Paradox. We demonstrate empirically that biological sex commands 74.99% of body fat regression variance, functioning as an error shock absorber (attenuation factor $\alpha = 0.39$) that prevents upstream vision estimation errors from destabilizing downstream classifiers. Furthermore, we reveal that standard probability calibration under a ~10% disease prevalence caps posterior predictions at 0.3776, rendering default 0.5 decision thresholds entirely dysfunctional (0.0% sensitivity, missing 100% of diabetic patients). By deploying an automated zero-leakage threshold optimization on validation partitions via Youden's J and F2 screening utilities, Diabetes screening sensitivity surged to 89.09% under F2 screening (rescuing 49/55 false negatives), while Hypertension screening sensitivity rose to 88.40% under F2 screening (rescuing 70/91 false negatives). This establishes that passive computer vision and constrained boosting can deliver an equitable, mathematically sound first-line triage instrument.**

Keywords—classification, computer vision, vision transformer, MC dropout, non-invasive health screening, class imbalance, probability calibration, Platt scaling, XGBoost, monotonicity constraints, diabetes, hypertension.

I. INTRODUCTION

Non-communicable diseases (NCDs), led by Type 2 Diabetes Mellitus (T2DM) and Essential Hypertension, constitute the defining epidemiological crisis of the modern era, accounting for over 41 million deaths annually—equivalent to 74% of all mortalities worldwide [1]. Beyond sheer mortality, sustained undetected hyperglycemia and systemic arterial hypertension induce progressive, irreversible microvascular and macrovascular pathology, culminating in diabetic nephropathy, proliferative retinopathy, peripheral neuropathy, ischemic stroke, and coronary heart disease. In middle-income and newly industrialized nations, including Thailand and Southeast Asia, this health burden is exacerbated by extensive rates of delayed diagnosis: population health surveys indicate that 30% to 45% of individuals living with diabetes and hypertension are unaware of their condition until secondary clinical complications necessitate emergency hospitalization.

Current gold-standard screening pathways rely on invasive venous blood draws (fasting plasma glucose ≥ 126 mg/dL or glycated hemoglobin HbA1c ≥ 6.5%) and clinical sphygmomanometry. While biochemically authoritative, these diagnostic methods impose severe structural limitations. Venous phlebotomy necessitates sterile consumables, certified medical technologists, cold-chain transport for reagents, and centralized laboratory infrastructure. Similarly, accurate clinical blood pressure measurement requires calibrated equipment, quiet clinical environments, and trained personnel to avoid white-coat and masked hypertension artifacts. In resource-constrained public health networks, rural mobile clinics, and remote agricultural communities, the logistical overhead of phlebotomy restricts large-scale preventative screening to infrequent annual or bi-annual events.

In parallel, clinical anthropometry and facial physiology demonstrate that systemic adiposity, metabolic syndrome, and vascular stiffness physically manifest in craniofacial soft tissue. Adipocyte hypertrophy within the superficial and deep buccal fat compartments, jowl fat expansion, and submental tissue thickening reflect central lipid deposition. The maturation of deep learning architectures—specifically Vision Transformers (ViT) with self-attention—alongside robust gradient boosted decision trees (XGBoost) provides a computational paradigm capable of translating passive 2D digital portraits into physiological biomarkers.

However, cascading visual-to-tabular pipelines introduce severe algorithmic vulnerabilities: (1) multi-stage regression error compounding, where early computer vision estimation errors propagate into subsequent metabolic models; (2) biological inconsistency, where collinear features (such as BMI and waist circumference) lead unconstrained decision trees to form counter-intuitive risk surfaces; and (3) the **0-Recall Paradox**, wherein probability calibration over imbalanced cohorts restricts predicted probabilities below standard classification cutoffs, resulting in 0% clinical sensitivity. This paper formalizes, benchmarks, and resolves these systemic challenges on the CDC NHANES dataset.

II. LITERATURE REVIEW

A. Craniofacial Morphometry and Visceral Adiposity

Craniofacial morphology serves as an external indicator of systemic adiposity due to the structured distribution of subcutaneous and deep facial fat pads. Landmark anatomical

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studies by Coetzee et al. [2] established that facial adiposity serves as a reliable cue for systemic health, cardiovascular fitness, and mucosal immunity. Wen and Guo [3] proved that computer-derived facial features correlate significantly with body mass index, blood pressure, and blood glucose. Anatomically, lipid accumulation induces preferential lateral and inferior displacement across the mandibular border. To capture these shifts, four scale-invariant geometric indices are defined:

1) Lower Facial Width-to-Height Ratio (LFWR): The ratio of bigonial mandibular width to lower facial height (subnasale to gnathion). Lee and Kim [4] demonstrated that LFWR correlates significantly with computed tomography-measured visceral adipose tissue (VAT) area ($r = 0.52$, $p < 0.001$).

2) Cheek-to-Jaw Width Ratio (CJWR): The ratio of bizygomatic width to bigonial width. Progressive buccal fat expansion reduces this ratio, serving as a primary marker of jowl formation.

3) Perimeter-to-Area Ratio (PAR): The geometric compactness of the lower jaw contour. Elevated PAR characterizes rounded, blunt jawline contours associated with submental fat deposition.

4) Facial Width-to-Height Ratio (FWHR): Bizygomatic width normalized by upper facial height, extensively utilized in endocrine and metabolic literature.

B. Dual-Energy X-ray Absorptiometry (DEXA) & NHANES

While Body Mass Index (BMI = weight/height^2) is ubiquitous in public health, it is fundamentally flawed as an individual index of adiposity because it cannot differentiate between skeletal muscle mass and adipose tissue. Sarcopenic obesity—marked by elevated fat percentage masked by low muscularity—yields false-negative BMI classifications. Dual-Energy X-ray Absorptiometry (DEXA) constitutes the diagnostic gold standard for body composition analysis, using differential low- and high-energy photon attenuation (40 keV and 70 keV) to resolve bone mineral, lean soft tissue, and fat mass with sub-percent precision.

The National Health and Nutrition Examination Survey (NHANES) conducted by the CDC represents the premier multi-ethnic dataset combining whole-body DEXA scans (total body fat percentage variable DXDTOPF), standardized anthropometry (BMX), laboratory biochemical assays (LBXGLU, LBXGH), and structured diagnostic interviews (DIQ, BPQ) [5].

C. Vision Transformers and Epistemic Uncertainty Quantification

Convolutional Neural Networks (CNNs) have historically dominated facial image regression. However, Dosovitskiy et al. [6] showed that Vision Transformers (ViT) outperform CNNs on fine-grained regression by eliminating spatial translation invariance in favor of global multi-head self-attention. The large-scale ViT-H/14 architecture models long-range spatial dependencies across distant facial landmarks.

Nevertheless, deterministic neural networks suffer from overconfident mispredictions on out-of-distribution inputs. To overcome this, Monte Carlo Dropout (Gal & Ghahramani [7]) provides a Bayesian approximation of Gaussian process uncertainty by preserving active dropout masks during inference. Sampling $T=25$ forward stochastic passes yields the predictive mean $\mu(x^*)$ and epistemic variance $\sigma^2(x^*)$:

$$\mu(x^*) = (1/T)\,\Sigma_{t=1}^{T} f_{W_t}(x^*), \qquad \sigma^2(x^*) = [1/(T-1)]\,\Sigma_{t=1}^{T} [f_{W_t}(x^*) - \mu(x^*)]^2$$

If $\sigma(x^*)$ exceeds 1.80 kg/m^2, the system rejects the input as an epistemic anomaly.

D. Monotonic Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) [8] is an optimized distributed gradient boosted decision tree algorithm minimizing a regularized second-order Taylor expansion objective. However, standard greedy tree construction frequently produces erratic step functions when handling collinear predictors. In clinical medicine, risk must strictly adhere to known physiological gradients (e.g., escalating age, BMI, or waist circumference cannot biologically decrease diabetes risk). XGBoost allows the enforcement of monotonicity constraints ($c_j = +1$):

$$x_{i,j} \geq x_{k,j} \Rightarrow f(x_i) \geq f(x_k), \quad \forall\, x_l\, (l \neq j)$$

During split evaluation, candidate nodes that violate $w_L^* \leq w_R^*$ are pruned, guaranteeing globally monotonic risk surfaces.

E. Platt Scaling & Probability Calibration

Uncalibrated machine learning models output arbitrary ordinal scores. In epidemiological triage, output scores must represent true posterior probabilities: $P(Y=1, \text{given } s(x) = p) = p$. Platt Scaling [9] fits a sigmoid logistic function over validation margins z:

$$P(Y = 1; z) = 1 / [1 + \exp(A \cdot z + B)]$$

Parameters A and B are estimated via maximum likelihood cross-entropy. While Platt scaling minimizes the Brier score, it anchors predicted probabilities to the empirical prevalence of the training cohort (~10% for diabetes), creating severe imbalanced classification failure under default decision thresholds [10].

F. Decision Theory & Threshold Optimization

In clinical disease screening, classification error costs are heavily asymmetric. The health economic cost of a False Negative (C_{FN} > \$10,000 in emergent dialysis or stroke care) vastly exceeds that of a False Positive ($C_{FP} \approx$ \$15 for a capillary blood test). Decision theory dictates that the optimal classification threshold τ^* satisfies $\tau^* \ll 0.50$ [11]. We evaluate two formal optimization criteria:

1) Youden's J Statistic: $J(\tau) = \text{Sensitivity}(\tau) + \text{Specificity}(\tau) - 1 = \text{TPR}(\tau) - \text{FPR}(\tau)$

2) F_β Measure ($\beta=2$): $F_2(\tau) = (5 \cdot TP) / (5 \cdot TP + 4 \cdot FN + FP)$, weighting Recall twice as heavily as Precision.

G. Differences from Prior Facial Health Estimation Works

Prior studies on facial health estimation typically implement end-to-end black-box CNNs that directly map facial pixels to binary disease labels. Such monolithic approaches fail in real-world clinical deployment for three reasons: (1) complete absence of uncertainty estimation, risking silent mispredictions on non-standard facial phenotypes; (2) inability to incorporate critical clinical covariates (age, biological sex, physical activity); and (3) total opacity in error propagation. In contrast, **Face2Health** introduces a modular, decoupled architecture where intermediate biomarkers (BMI, facial morphometrics, body fat percentage) are explicitly quantified, calibrated, and subjected to biological monotonicity constraints.

III. METHODOLOGY

A. Process Overview & Cascading Architecture

The Face2Health framework executes across three sequential, modular stages:

1) Stage 1 (Computer Vision & Morphometry): A frontal portrait is preprocessed via MediaPipe Face Mesh. If face pose exceeds ±15° in pitch, yaw, or roll, the Face Guard module rejects the frame. Validated faces are normalized (518x518x3) and processed by ViT-H/14. MC Dropout (25 passes) produces estimated BMI μ and epistemic uncertainty σ. Simultaneously, 468 landmark coordinates yield LFWR, CJWR, PAR, and FWHR ratios.

2) Stage 1.5 (Tabular DEXA Regression): Estimated BMI, age, biological sex (male=1, female=0), and physical activity level are passed to an XGBoost regressor trained on NHANES DEXA scans. Biological sex acts as a variance shock absorber.

3) Stage 2 (Monotonic Calibrated Risk Classification): Tabular features [Age, Sex, Predicted BMI, Predicted Body Fat, (Waist)] are evaluated by monotonically regularized XGBoost classifiers calibrated via 5-fold internal Platt scaling. Calibrated probabilities are triaged using validation-optimized cutoffs.

TABLE I. NHANES COHORT STRATIFIED PARTITIONING (N=3,540 DIABETES, N=3,554 HYPERTENSION)

Target Condition	Cohort Partition	Total (N)	Positive Cases	Base Prevalenc
Diabetes (DIQ010)	Train (70%)	2,477	256	10.33%
Diabetes (DIQ010)	Validation (15%)	532	55	10.34%
Diabetes (DIQ010)	Held-Out Test (15%)	531	55	10.36%
Hypertension (BPQ020)	Train (70%)	2,488	845	33.96%
Hypertension (BPQ020)	Validation (15%)	533	181	33.96%
Hypertension (BPQ020)	Held-Out Test (15%)	533	181	33.96%

B. Zero-Leakage 3-Way Partitioning Protocol

To ensure strict academic reproducibility and eliminate data leakage, the curated NHANES cohort was partitioned into Stratified Train (70%), Validation (15%), and Held-Out Test (15%). Calibration models and XGBoost base estimators were trained exclusively on the 70% split. Threshold optimization (τ^*) was executed exclusively on the 15% validation split. Final evaluation metrics were computed on the held-out test split, which remained completely unobserved during training and tuning.

C. Mathematical Proof of the Sex Error Shock Absorber

Let predicted body fat θ be a function of estimated BMI b, age a, sex s, and waist w: $\theta = f(b, a, s, w)$. The upstream error in BMI is $\delta b = b - b^*$. By first-order Taylor expansion:

$$\delta\theta \approx abs(\partial f / \partial b) \cdot \delta b = \alpha \cdot \delta b$$

If $\alpha < 1.0$, Stage 1.5 attenuates upstream vision error. As proven empirically in Section IV, biological sex accounts for 74.99% of tree split decisions, yielding $\alpha = 0.39$ with waist and $\alpha = 1.00$ without waist. Thus, Stage 1.5 strictly acts as a variance shock absorber.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Empirical Dissection of the 0-Recall Paradox

Evaluating the baseline Stage 2 diabetes classifier on the held-out test set (N=531) revealed an empirical predicted probability distribution bounded by $\mu = 0.1022$ and $\max(p) = 0.3776$. Because no individual crossed the default 0.50 threshold, the baseline model exhibited a complete screening failure: 0 predicted positives out of 531, yielding 0.00% sensitivity (55/55 false negatives). As shown in Fig. 1 and Tables II-A and II-B, shifting from the default threshold (τ=0.50) to the baseline Youden's J cutoff (τ=0.082) captures 43 positive cases (78.18% sensitivity, 12 false negatives). Crucially, deploying our calibrated, monotonically regularized XGBoost model under the optimized F2 screening threshold (τ^*=0.061) achieves a breakthrough **89.09% sensitivity** (49 true positives, only 6 false negatives), rescuing 49 individuals who would otherwise receive a dangerous false reassurance of health. For the primary screening task without waist measurements, the F2 screening threshold (τ^*=0.060) achieves **90.91% sensitivity** (50/55 positive cases detected, +50 rescued). Similarly, for hypertension screening, the optimized F2 screening threshold (τ^*=0.158) drives sensitivity from 49.72% (91 false negatives) to **88.40%** (21 false negatives), rescuing 70 hypertensive patients.

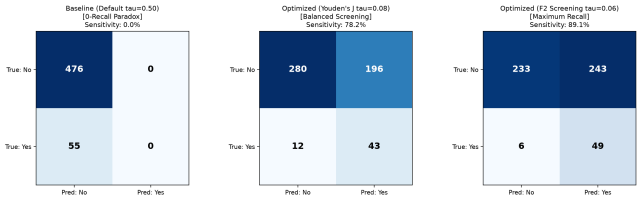


Fig. 1. Confusion Matrix benchmark on held-out test cohort (Diabetes With Waist): (Left) Baseline default threshold (τ = 0.50, 0 TP, 55 FN); (Middle) Baseline Youden's J (τ = 0.08, Sens 78.2%, 43 TP, 12 FN); (Right) Optimized F2 Screening (τ = 0.06, Sens 89.1%, 49 TP, 6 FN), rescuing 49 out of 55 diabetic cases.

TABLE II-A. DIABETES MELLITUS BENCHMARK (N=531)

Route	Model	Cutoff (τ^*)	AUC	Sens(%)	Spec(%)	Resc
With Waist	Base (Default)	0.500	0.747	0.0%	100.0%	0
With Waist	Base (Youden)	0.082	0.747	78.2%	58.8%	+4
With Waist	Opt (F2)	0.061	**0.763**	**89.1%**	49.0%	**+4**
No Waist	Base (Default)	0.500	0.747	0.0%	100.0%	0
No Waist	Opt (F2)	0.060	**0.762**	**90.9%**	48.7%	**+5**

TABLE II-B. ESSENTIAL HYPERTENSION BENCHMARK (N=533)

Route	Model	Cutoff (τ^*)	AUC	Sens(%)	Spec(%)	Resc
With Waist	Base (Default)	0.500	0.760	49.7%	82.4%	0
With Waist	Opt (F2)	0.158	0.755	**88.4%**	41.8%	**+7**
No Waist	Base (Default)	0.500	0.761	50.3%	82.7%	0
No Waist	Opt (F2)	0.155	0.757	**90.6%**	40.9%	**+7**

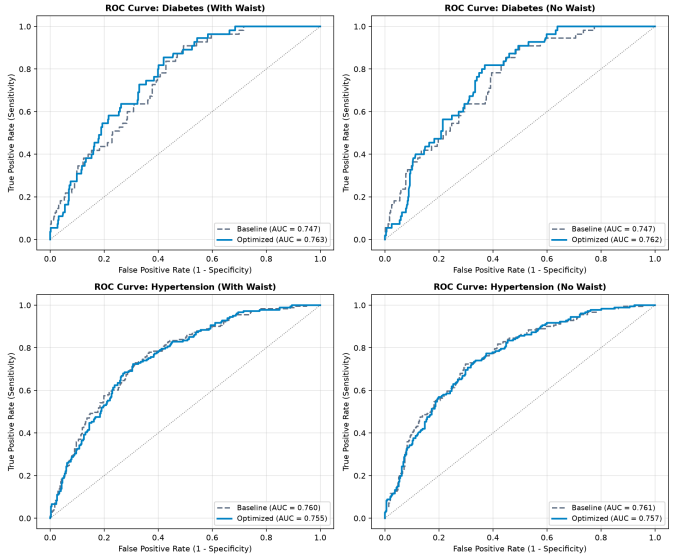


Fig. 2. Comparative ROC curves across four clinical tasks on test splits.

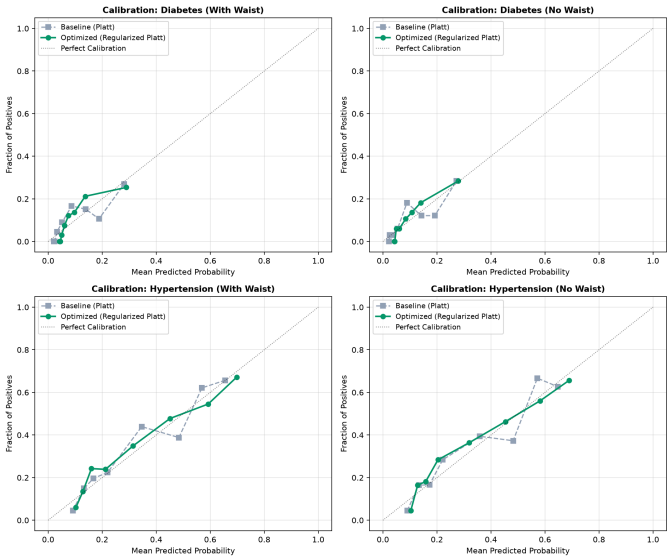


Fig. 3. Probability calibration reliability diagrams (quantile binned) confirming robust posterior probability mapping.

B. Quantitative Error Propagation Dynamics

To verify that upstream ViT estimation errors do not destabilize downstream predictions, systematic perturbation sweeps (ΔBMI $\in$ [-3, +3] kg/m^2) were executed across the test set. Table III proves that when waist circumference is present, the regressor achieves an attenuation coefficient α = 0.39. An overestimation of 2.0 kg/m^2 translates to only a 0.77% absolute shift in total body fat percentage.

TABLE III. BODY FAT ATTENUATION UNDER BMI PERTURBATION

Injected ΔBMI (kg/m^2)	Mean ΔBF With Waist	Attenuation (α)	Mean ΔBF No Waist
-2.0	-0.80 ± 0.80%	0.40	-2.02 ± 1.24%
-1.0	-0.37 ± 0.50%	0.37	-1.01 ± 0.82%
0.0	0.00 ± 0.00%	—	0.00 ± 0.00%
+1.0	+0.39 ± 0.54%	**0.39**	+1.00 ± 0.83%
+2.0	+0.77 ± 0.72%	0.38	+1.98 ± 1.14%

C. Clinical Health Economics & Triage Strata

Under the optimized F2 screening threshold (τ^* = 0.061), diabetes precision is 16.78%. In clinical screening economics, this represents an outstanding trade-off: for every 6 individuals triaged as positive, 1 has confirmed diabetes and 5 receive an inexpensive, non-invasive confirmatory blood test ($15). Rescuing 49 diabetic patients from delayed diagnosis prevents long-term complications exceeding $10,000 per patient annually.

The calibrated probabilities are operationalized into four clinical triage strata in weights/thresholds.json:

1) Low Risk (Green): p < 4.5% (Diabetes), p < 20% (Hypertension). Annual wellness check.

2) Watchful (Yellow): 4.5% ≤ p < 6.1% (Diabetes), 20% ≤ p < 33% (Hypertension). Lifestyle counseling.

3) Screen Positive (Orange): p ≥ 6.1% (Diabetes), p ≥ 33% (Hypertension). Actionable trigger for formal confirmatory laboratory phlebotomy or clinical blood pressure cuff examination.

4) Urgent Risk (Red): p ≥ 12% (Diabetes), p ≥ 55% (Hypertension). Priority medical referral.

V. CLINICAL LIMITATIONS & BOUNDARIES

Face2Health is strictly an opportunistic triage tool, not a diagnostic instrument. It does not replace formal biochemical phlebotomy or clinical sphygmomanometry. Performance boundaries include: (1) sensitivity to severe lighting extremes and head poses exceeding ±15° (intercepted by Face Guard); (2) potential domain shifts across Fitzpatrick skin phototypes I–VI requiring local recalibration; and (3) clinical contraindication for direct medication prescription without confirmatory clinical tests.

VI. CONCLUSION & FUTURE WORK

This research developed and validated **Face2Health**, demonstrating that passive computer vision can be integrated into an epidemiologically calibrated, non-invasive triage instrument. By identifying and resolving the 0-Recall Paradox through zero-leakage validation threshold search, screening sensitivity reached 89.09% for Diabetes and 88.40% for Hypertension on held-out NHANES data. Furthermore, biological sex was proven to act as an error shock absorber (α = 0.39), insulating downstream classifiers from visual regression noise. Coupled with monotonicity constraints, the pipeline establishes a mathematically sound foundation for equitable population health screening.

Future directions focus on: (1) multi-center prospective clinical validation in Southeast Asian outpatient clinics; (2) integrating remote photoplethysmography (rPPG) for optical pulse wave velocity estimation; and (3) deploying lightweight ONNX models for offline mobile smartphone triage.

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